LESSON 01 · MISSION GUIDE

Mistress Pell's Ravens

Numbers, text and formatting: the bread and butter of every treasury.

From **Mistress Pell, Paymaster of the Guilds**
To **Jen, Keeper of the Ledger**
By raven **Seven ravens, in fact (help)**

Morning Jen! Hope the gate guard let you in. Quick word from the Paymaster's desk:

The Grimoire failed at the full moon with "Run-time error '13': Type mismatch", which is Dalvric's way of saying hello from beyond the western sea. I've got seven ravens that need answering: pay maths, a messy copy of the Roll, pay scrolls the Queen thinks look "a bit 1019", and numbers the Scriptorium insists are text. Oh, and Her Majesty wants the pay gap "in one sentence" for the Privy Council on Thursday.

Orrin says Python can do all of this. Orrin also said the Scriptorium would stop smudging by Tuesday. No pressure! — P.

What you'll be able to do after today

- [ ] Store values in **variables** and calculate with them: `+ - * /`, plus `//`, `%` and `**`
- [ ] Clean up messy text: trim spaces, fix capitals, replace characters
- [ ] Turn text like `"32,500.00 cr"` into a number you can calculate with, and back again
- [ ] Format output like Excel number formats: `2,708.33 cr`, `12.5%`, neat columns
- [ ] Ask yes/no questions with `True` and `False`
- [ ] Cut text apart like `LEFT`, `RIGHT` and `MID`, but counting from 0

Why this matters at work

- **Every scroll arrives as text.** CSVs, the Scriptorium, the Clockworks: all text, even the salaries. Converting it properly is step one of every analysis.
- **Pay scrolls and reports are formatted numbers.** If you can make one scroll line look right, you can make 4,000 of them look right.
- **The pay gap is just arithmetic.** Exercise 7 is the same formula the final project uses on the whole Roll of the Realm.

Four big ideas

1. No `Dim`	`salary = 32500` creates the variable. Python works out the type itself.
2. Everything has a type	`int` (42), `float` (37.5), `str` ("Maren"), `bool` (True). `type(x)` tells you which.
3. Python never guesses	Excel turns "5" into 5 behind your back. Python refuses: `"5" + 3` is an error. You convert on purpose with `int()`, `float()`, `str()`.
4. Counting starts at 0	The first letter of `"Maren"` is `"Maren"[0]`. Excel would say position 1.

Three VBA habits that will bite you

- `^` is **not** "to the power of". Use `**`. (`2 ^ 10` is 8. Don't ask.)
- `=` stores a value, `==` compares. VBA uses `=` for both.
- Names are **case-sensitive**: `Salary` and `salary` are two different variables.

How today works

15 MIN
Lecture
Lars demos everything live, mistakes included

60 MIN
Exercise sheet
Seven ravens to answer. You write every file yourself

15 MIN
Grimoire, page 1
Three bugs. The Royal Auditors are waiting

The ravens

#	FROM	RAVEN	YOU'LL USE
1	Mistress Pell	Pay maths	`/ // % ** round()`
2	Orrin	Clean up the Scriptorium's copy	`.strip() .title() .lower() .replace()`
3	The Queen	A pay scroll that isn't "a bit 1019"	f-strings: `f"{x:,.2f} cr"`
4	Mistress Pell	Text that should be numbers	`float() int() type()`
5	Lady Siv	Yes or no	`== <= and or not`
6	Orrin	LEFT, RIGHT, MID	`text[:3] text[-4:] .find()`
7 ⭐	The Queen	The pay gap in one sentence	all of the above

Your three documents

GUIDE.pdf	This one.
EXERCISES.pdf	The exercise sheet: what to write, and the expected output to check against.
CHEATSHEET.pdf	Everything from today on one page. Keep it open while you work.

You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if you're feeling brave), each printing the expected output
- [ ] Dalvric's page prints the expected two lines, with one commit per bug
- [ ] Everything is pushed to `lesson-01` and you've opened a pull request



Next lesson: The Midwinter Bonus. Lady Siv has 200 subjects of the realm and a set of bonus rules the Queen changes every rest day. You'll teach Python to make decisions (`if`) and to repeat things without copy-paste (`for`).